Patellar Tendon or Semitendinosus Tendon Autografts for Anterior Cruciate Ligament Reconstruction

A Prospective, Randomized Study With a 7-Year Follow-up

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Background: The aim of the study was to compare the results after arthroscopic anterior cruciate ligament (ACL) reconstruction using central-third bone–patellar tendon–bone (BTB) autografts and triple/quadruple semitendinosus (ST) autografts.

Hypothesis: In the long-term, ACL reconstruction using BTB autografts will render more donor-site problems than ST autografts.

Study Design: Randomized controlled trial; Level of evidence, 1.

Methods: A randomized series of 71 patients (22 women and 49 men) with a unilateral ACL rupture who underwent reconstructive surgery were included in the study. The BTB graft was used in 34 patients (BTB group) and the ST-tendon graft was used in 37 patients (ST group). The patients were examined a median of 86 months (range, 68 to 114 months) after the reconstruction.

Results: Sixty-eight of 71 patients (96%) were examined at follow-up. The clinical assessments at follow-up revealed no significant differences between the BTB group and the ST group in terms of the Lysholm score, Tegner activity level, International Knee Documentation Committee evaluation system, 1-legged hop test, KT-1000 arthrometer laxity measurements, manual Lachman test, and range of motion. A significant improvement was seen in both groups compared with the preoperative values in terms of most clinical assessments. Donor-site morbidity in the form of knee-walking ability, kneeling ability, and area of disturbed anterior knee sensitivity revealed no significant differences between the groups.

Conclusion: Seven years after ACL reconstruction, the subjective and objective outcomes were similar after using the central-third BTB autograft and triple/quadruple ST autograft. Furthermore, no difference in terms of donor-site morbidity was found between the 2 groups.

Keywords: anterior cruciate ligament; patellar tendon; semitendinosus; autograft; surgery; randomized; prospective

Rupture of the anterior cruciate ligament (ACL) is a common injury and intra-articular ACL reconstruction using autologous materials is widely used. The results are reported as good and reproducible in several studies using different autologous grafts harvested from around the knee region.[4,5,9,10,22] The use of the central third of the patellar tendon with bone blocks at either end (bone–patellar tendon–bone, or BTB) as an autograft for arthroscopic reconstruction of the ruptured ligament has become a standard procedure during the past decade, sometimes even called the "gold standard."[21,23] However, well-known problems have been reported in conjunction with this method, especially in terms of donor-site morbidity, such as anterior knee pain, disturbances in knee sensitivity, and kneeling discomfort.[18,31,34,39,48] It has been suggested that by obtaining

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full extension of the knee postoperatively, anterior knee pain can be prevented.[34,50] A frequently used alternative to the BTB autograft is the use of semitendinosus (ST) or ST/gracilis autografts. In a meta-analysis of available studies, Freedman et al[20] reported that reconstruction using BTB autografts has a lower rate of graft failure, less objective knee laxity, and increased patient satisfaction; however, it results in an increased rate of anterior knee pain compared with the use of hamstring tendon autografts. In another meta-analysis, Yunes et al[55] found a trend toward improved stability using the BTB autograft compared with the hamstring tendon autograft and no significant differences in either complications or failure rate. Recently, in a meta-analysis of all prospective randomized clinical trials comparing BTB and ST autografts with a minimum of 2 years of follow-up, Goldblatt et al[24] found 11 studies fulfilling the criteria for inclusion. The maximum follow-up time in this meta-analysis was a mean of 52 months. In their analyses, the BTB graft was found to be more likely to result in a normal Lachman test, normal pivot-shift test, better KT-1000 arthrometer values, and fewer patients with loss of flexion. The ST graft, on the other hand, resulted in a lower incidence of patellofemoral crepitance, less kneeling pain, and fewer patients with loss of extension. In an extensive review, Beynnon et al[5] found that reconstruction using the 4-strand hamstring graft results in similar clinical and functional outcomes compared with the BTB graft.

Although several prospective, randomized, short-term and medium-term studies have been published,[2,6,16-18,30,44,48] to our knowledge, few randomized outcome studies with more than a 5-year follow-up can be found in the literature. Ibrahim et al[29] examined only male patients at a mean of 81 months. However, only 77% of the included patients were examined at follow-up. They found no differences in terms of subjective complaints or objective laxity measurements using the KT-1000 arthrometer. There were, however, more patients with patellofemoral problems and loss of knee motion after using the BTB graft. Matsumoto et al[42] found a reduced risk of problems at the graft harvest site using the bone-hamstring-bone graft compared with the BTB autograft, but the results in the remaining clinical parameters that were tested a minimum of 5 years postoperatively were comparable.

The aim of the present study was to compare the outcome of ACL reconstruction after using the subcutaneously harvested BTB graft with the use of a triple/quadruple ST-tendon graft in a long-term follow-up. Our hypothesis was that ACL reconstruction using either the BTB graft or the ST graft would render similar clinical results in the long term, apart from the presence of more donor-site problems after using the BTB graft.

MATERIALS AND METHODS

Patients

Patients with a symptomatic unilateral chronic ACL rupture were prospectively randomized for reconstruction using either an ipsilateral BTB graft or an ipsilateral triple/quadruple ST graft. The randomization was carried out using closed envelopes. The aim was to randomize 80 patients. However, due to the time factor, the study was terminated after randomizing 71 patients. In 34 patients, the central third of the patellar tendon was used as a graft (BTB group), and in 37 patients, the ST was used in the form of a triple (n = 14) or quadruple (n = 23) graft (ST Group).[39] The groups were comparable in terms of gender, age, preinjury Tegner activity level, and time between the injury and the index operation (Table 1). The index operations were performed between September 1995 and October 1997.

TABLE 1
Preoperative Data[a]

Variable	BTB Group	ST Group	Significance
Number of patients	34	37	
Age in years (range)	28 (14-49)	29 (15-59)	n.s. (P = .8)
Gender female/male	11/23	11/26	n.s. (P = .7)
Preinjury Tegner activity level (range)	9 (3-9)	9 (5-10)	n.s. (P = .8)
Time in months between the injury and index operation (range)	30 (2-252)	37 (3-360)	n.s. (P = .4)

[a]BTB, bone–patellar tendon–bone; ST, semitendinosus; n.s., not significant.

Surgical Technique

One senior surgeon, who was experienced in using both techniques, performed all the reconstructions. In the BTB group, the arthroscopic transtibial technique and interference screw fixation[38] were used during the index procedure. The midthird of the patellar tendon was harvested through 2 approximately 25-mm-long vertical incisions, one over the apex of the patella and the other just above the tibial tubercle. The graft was tunneled subcutaneously under the paratenon with the aim of protecting the infrapatellar branches of the saphenous nerve and leaving the major part of the paratenon intact, as previously described by Kartus et al.[32] This is done theoretically to minimize the area of disturbed sensitivity. The defects in the patella and the proximal tibia were not bone-grafted.[7] The proximal bone block was sized at 9 mm and the distal bone block at 10 mm. The bone tunnels were prepared in a standard transtibial fashion. A 7-mm and a 9-mm interference screw (Acufex Microsurgical, Mansfield, Mass) were used on the femoral and the tibial side, respectively (Figure 1).

In the ST group, the graft was harvested through a 3-cm oblique incision over the pes anserinus. The tendons were palpated and the sartorius fascia was incised parallel to the fibers of the fascia just above the thicker and more distally inserted ST tendon. After the vinculae had been cut under visual control, the tendon was harvested with a semiblunt, semicircular open tendon stripper (Acufex Microsurgical). The tendon was prepared for a triple or quadruple graft, depending on the length of the tendon. The minimum

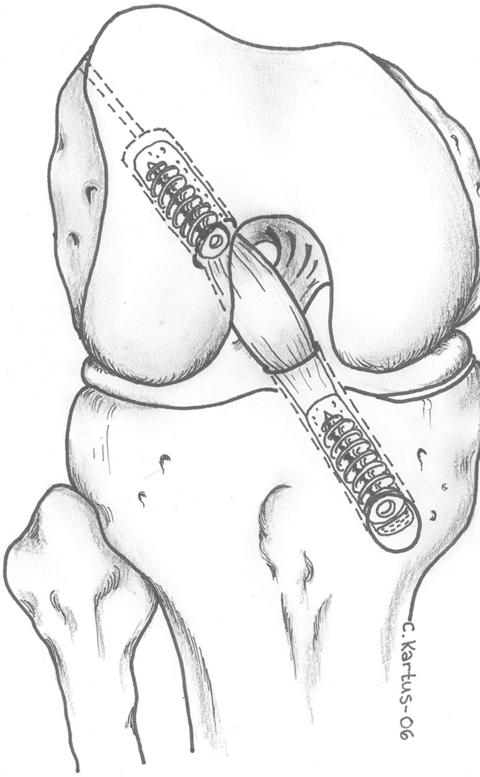

Figure 1. In the bone–patellar tendon–bone group, a 7-mm and a 9-mm interference screw were used on the femoral and tibial side, respectively. (©Catarina Kartus)

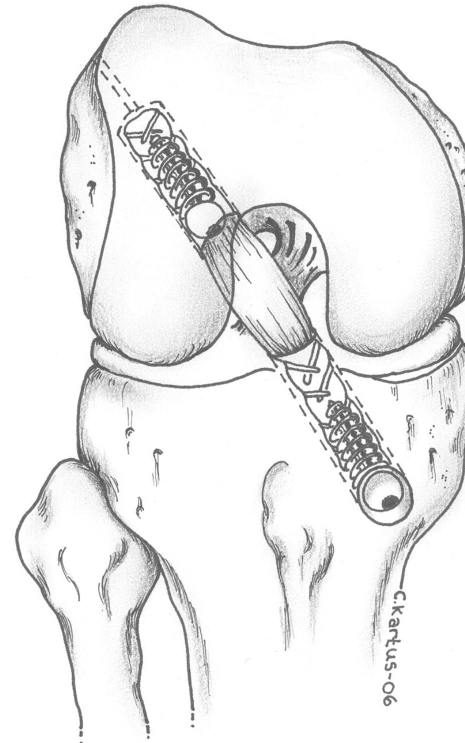

Figure 2. In the semitendinosus group, a 7-mm round-headed, soft-threaded interference screw was used on both the femoral and tibial sides. (©Catarina Kartus)

accepted length for the final graft was 7 cm. Two No. 5 nonresorbable Ticron sutures (Sherwood Medical, St Louis, Mo) were used as the lead sutures at the distal and proximal ends. Resorbable No. 1 Vicryl (Ethicon GmbH & Co KG, Norderstedt, Germany) sutures were used for the modified baseball stitches at the distal and proximal ends of the ST graft. The bone tunnels were prepared in the same transtibial fashion as in the BTB group. A 7-mm soft-threaded interference screw (RCI, Smith & Nephew, Andover, Mass) was used on both the femoral and tibial side[11] (Figure 2). After the femoral screw had been inserted, firm traction was applied to the graft during the insertion of the tibial screw, with the knee in hyperextension, in both the BTB group and the ST group.

Clinical Assessments and Follow-up

Two independent physiotherapists who were not involved in the rehabilitation performed all the preoperative and postoperative assessments. The manual Lachman test (graded as 0, +1, +2, +3) and the KT-1000 arthrometer (MEDmetric Corp., San Diego, California) were used for the assessment of laxity.[12] The Lysholm score,[52] Tegner activity level,[52] and International Knee Documentation Committee (IKDC) evaluation system[26] were used as scoring systems. The KT-1000 arthrometer anterior side-to-side difference was measured at 89 N preoperatively and at 89 N and 134 N at follow-up. The Lysholm score was self-administered according to Höher et al.[27] Range of motion (ROM) was measured to the nearest 5° using a goniometer. Measurements of disturbances in sensitivity in the anterior knee region were performed by palpation and measured in cm^2. The patients were classified as having subjective anterior knee pain if they registered pain during stair-walking, sitting with the knee in 90° of flexion, or during or after activity. The classification of kneeling discomfort was performed using the knee-walking and the kneeling test.[37] The 1-legged-hop test[53] was used to evaluate the functional performance.

Rehabilitation

All the patients were rehabilitated according to the same accelerated protocol, permitting immediate full weight-bearing and full ROM.[49] No rehabilitation brace was used.[8,28,36] Closed kinetic chain exercises were started immediately postoperatively. Terminal extension with an external load other than the weight of the operated leg was not permitted during the first 6 postoperative weeks. Running was permitted at 3 months and contact sports at 6 months at the earliest, provided that the patient had regained full subjective functional stability.

TABLE 2
Causes of Additional Surgery During the 7-Year Follow-up Period[a]

Additional Surgery	BTB Group	ST Group	Total
None	24	29	53
Meniscus problem	2	1	3
Screw problem	2	1	3
Screw and meniscus problems	1	0	1
ACL reinjury	1	2	3
Reinjury, other than ACL	1	0	1
Other	1	3	4
Total	32	36	68

[a]Three patients were lost to follow-up. BTB, bone–patellar tendon–bone; ST, semitendinosus; ACL, anterior cruciate ligament.

TABLE 3
Lysholm Score, Tegner Activity Level, and 1-Legged Hop Test: Preoperative and Postoperative Data[a]

Outcome Measure	BTB Group	ST Group	Significance BTB vs ST
Lysholm Score			
Preoperative	70 points (14-95)	68 points (21-100)	n.s. (P = .5)
7-year follow-up	81 points (25-100)	90 points (50-100)	n.s. (P = .2)
Significance preop vs postop	P = .001	P < .0001	
Missing values postop	1	2	
Tegner Activity Level			
Preoperative	3 (1-8)	4 (2-9)	n.s. (P = .6)
7-year follow-up	5 (0-7)	6 (2-9)	n.s. (P = .2)
Significance preop vs postop	P = .001	P = .0001	
Missing values postop	0	0	
1-Legged Hop Test			
Preoperative	84% (0-111)	81% (0-108)	n.s. (P = .4)
7-year follow-up	96% (0-119)	92% (0-110)	n.s. (P = .1)
Significance preop vs postop	P = .001	P = .004	
Missing values postop	1	2	

[a]Median values (range) are presented. BTB, bone–patellar tendon–bone; ST, semitendinosus; n.s., not significant.

Statistical Methods

Median (range) values are presented, except for the anterior KT-1000 arthrometer laxity measurements for which mean (range) values are presented. The Wilcoxon signed rank test was used for comparisons of the preoperative and postoperative data within the groups. The Mann-Whitney U test was used to compare the variables between the groups. The χ^2 test was used to compare dichotomous variables. A P value of <.05 was considered statistically significant.

The primary variables in the study were the KT-1000 arthrometer laxity measurements and the anterior knee problems as measured with the knee-walking test. We expected a difference of at least 1 unit in the knee-walking test when planning the study. With a standard deviation of 1 unit in this test, a sample size of about 30 patients in each group estimates the power to >90%. Correspondingly, a difference of 1.5 mm with a standard deviation of 2 mm in terms of the KT-1000 arthrometer measurements estimates the power to >80%.

Ethics

The Human Ethics Committee of the medical faculty at Göteborg University approved the study.

RESULTS

Seventy-one patients (22 women and 49 men) were included in the study and 68 patients (96%) attended the follow-up at a median of 86 months (range, 68 to 114). One patient in each group was lost to follow-up because we were unable to locate them, and 1 patient in the BTB group who had suffered a traumatic graft rupture did not feel motivated to participate in the follow-up examination. Moreover, 1 patient in the BTB group and 2 in the ST group suffered a traumatic graft rupture during the follow-up period and were examined and reported as failures but excluded from the calculations. The majority of index injuries occurred during contact sports (69%) and noncontact sports (16%). Meniscal injuries treated before the index operation, at the index operation, or during the follow-up period were registered in 24 of 32 patients (75%) in the BTB group and 27 of 36 (75%) in the ST group (not significant, P = 1.0). Articular surface damage or localized degenerative changes were found in 4 patients in each group at the index operation. One patient in the BTB group suffered clinical signs of bacterial arthritis, but the cultures were negative. One patient in the ST group suffered the same symptoms but with positive cultures. Both patients healed after treatment with arthroscopic lavage and antibiotics. At follow-up, 63 patients (93%) displayed a normal contralateral side. Fifty-three patients (78%) had not undergone any additional surgery during the follow-up period. Meniscal problems and pain at the tibial hardware insertion site were the most common causes of additional surgery (Table 2).

At follow-up, there were no significant differences in terms of the Lysholm score, Tegner activity level, and 1-legged-hop test between the study groups. Both the BTB group and the ST group improved significantly between the preoperative assessment and follow-up (Table 3).

The KT-1000 arthrometer anterior side-to-side difference at follow-up did not demonstrate any differences between the study groups, either at 89 N or at 134 N. The side-to-side difference at 89 N displayed a significant decrease in the BTB Group (P = .004). In the ST group, the

TABLE 4
Laxity Assessments: Preoperative and 7-Year Follow-up Data[a]

Outcome Measure	BTB Group	ST Group	Significance BTB vs ST
KT-1000 Arthrometer Anterior Side-to-Side Difference			
Preoperative at 89 N	4.5 mm (–7-20)	4.3 mm (–3.5-24.5)	n.s. (P = .6)
7-year follow-up at 89 N	1.7 mm (–5-6)	2.6 mm (–3-9)	n.s. (P = .3)
7-year follow-up at 134 N	2.3 mm (–3.5-9)	2.7 mm (–4.5-9.5)	n.s. (P = .5)
Significance preop vs postop at 89 N	P = .004	n.s. (P = .2)	
Missing values postop	0	0	
Manual Lachman Test	(0) (+) (++) (+++)	(0) (+) (++) (+++)	
Preoperative	0 2 9 19	0 7 17 10	P = .006
7-year follow-up	8 21 1 0	7 23 1 1	n.s. (P = .6)
Significance preop vs postop	P < .0001	P < .0001	
Missing values postop	1	2	
IKDC			
Preoperative			
Normal or nearly normal	None	None	
Abnormal or severely abnormal	100%	100%	n.s. (P = 1.0)
Follow-up			
Normal or nearly normal	48%	50%	
Abnormal or severely abnormal	52%	50%	n.s. (P = .8)
Significance preop vs postop	P < .0001	P < .0001	
Missing values postop	1	3	

[a]KT-1000 arthrometer values are mean (range). BTB, bone–patellar tendon–bone; ST, semitendinosus; n.s., not significant; IKDC, International Knee Documentation Committee.

decrease did not reach statistical significance. At follow-up, the manual Lachman test revealed significantly reduced laxity in both groups compared with the preoperative assessment (Table 4).

There were no significant differences between the groups in terms of disturbance in anterior knee sensitivity and in terms of the ability to kneel and knee-walk at follow-up. Furthermore, no differences were found within the study groups when comparing the postoperative kneeling and knee-walking ability with the preoperative values. In the BTB group, 35% of patients and, in the ST Group, 18% of the patients classified the knee-walking test as difficult or impossible to perform at follow-up (not significant, P = .2) (Table 5). In the BTB group, 32% of the patients reported subjective anterior knee pain before reconstruction; at follow-up, subjective anterior knee pain was reported by 39%. The corresponding values in the ST group were 35% before reconstruction and 26% at follow-up (not significant, P = .3; BTB vs ST at follow-up).

No differences were found between the groups in terms of ROM postoperatively. Loss of extension or hyperextension of ≥5° compared with the noninjured contralateral side was registered in 23% of the patients in the BTB group and in 21% of the patients in the ST group (not significant, P = .4). The corresponding values for a flexion deficit of ≥5° were 39% in the BTB group and 35% in the ST group (not significant, P = .9) (Table 6).

DISCUSSION

The aim of this prospective randomized study was to report the medium- to long-term results after the use of a central third BTB graft and tripled or quadrupled ST graft for ACL reconstruction. The hypothesis was that ACL reconstruction using BTB autografts would render more donor-site problems than ST autografts. The hypothesis was thus rejected. Both grafts rendered similar improvements in function and laxity. Contrary to previous short-term studies, the presence of donor-site morbidity in the form of knee-walking and kneeling problems was comparable in the study groups.

The strength of this study was its prospective and randomized design, a follow-up rate of 96%, a power estimation of between 80% and 90% in terms of the primary variables, the fact that 1 surgeon experienced in both techniques performed all the reconstructions, and that 2 unbiased observers, who were not involved in the surgery or rehabilitation, performed all the preoperative and postoperative evaluations. Apart from the use of a soft-threaded type of interference screw in the ST group, the only variable that differed between the groups was the type of graft. All the other factors with an important effect on the final outcome, including the fixation technique and the rehabilitation protocol, were identical, thereby avoiding the bias of performance and detection.

TABLE 5
Knee-Walking Ability, Disturbance in Anterior Knee Sensitivity, and Kneeling Ability: Preoperative and Follow-up Data[a]

Outcome Measure	BTB Group	ST Group	Significance BTB vs ST
Knee-Walking Test			
Preoperative			
Normal	42%	44%	
Unpleasant	39%	41%	
Difficult	13%	9%	
Impossible	6%	6%	n.s. (*P* = .7)
Follow-up			
Normal	39%	50%	
Unpleasant	26%	32%	
Difficult	16%	9%	
Impossible	19%	9%	n.s. (*P* = .2)
Significance preop vs postop	n.s. (*P* = .1)	n.s. (*P* = 1.0)	
Missing values postop	0	0	
Disturbance in knee sensitivity (cm^2)			
Preoperative	0 (0-168)	0 (0-24)	n.s. (*P* = .08)
Follow-up	4 (0-299)	35 (0-282)	
Significance preop vs postop	*P* = .01	*P* = .0001	n.s. (*P* = .5)
Missing values postop	0	1	
Kneeling			
Preoperative			
Normal	58%	71%	
Unpleasant	39%	24%	
Difficult	0%	3%	
Impossible	3%	3%	n.s. (*P* = .4)
Follow-up			
Normal	52%	59%	
Unpleasant	26%	35%	
Difficult	16%	0%	
Impossible	6%	6%	n.s. (*P* = .3)
Significance preop vs postop	n.s. (*P* = .1)	n.s. (*P* = .3)	
Missing values postop	0	0	

[a]BTB, bone–patellar tendon–bone; ST, semitendinosus; n.s., not significant.

TABLE 6
Range of Motion: Preoperative and Follow-up Data[a]

Range of Motion	BTB Group	ST Group	Significance BTB vs ST
Preoperative			
Extension	–5° (–15-10)	–5° (–20-5)	n.s. (*P* = .1)
Flexion	145° (90-150)	145° (130-155)	*P* = .04
Follow-up			
Extension	–5° (–10-5)	–5° (–10-5)	n.s. (*P* = .4)
Flexion	140° (110-155)	140° (130-155)	n.s. (*P* = .9)
Significance			
Extension preop vs follow-up	n.s. (*P* = .5)	n.s. (*P* = .4)	
Flexion preop vs follow-up	n.s. (*P* = .2)	*P* = .003	
Missing values postop	0	0	

[a]Median values (range) are presented. BTB, bone–patellar tendon–bone; ST, semitendinosus; n.s., not significant.

Potential weaknesses of the study involve the lack of radiographic assessments as well as the fact that the KT-1000 arthrometer anterior side-to-side difference was only measured at 89 N preoperatively and at 89 N and 134 N at follow-up. For comparisons with other studies, it might have been better to use the manual maximum test and to perform radiographic examinations. For example, in a long-term study comparing hamstring tendon and BTB grafts for ACL reconstruction, Roe et al.[47] recently reported that radiographic degenerative changes were more common in the BTB group than in the hamstring tendon group.

The appropriate graft choice for ACL reconstruction still remains controversial. Numerous graft options and surgical techniques have been reported in the literature.[4,9,15,22,43] Two popular graft alternatives today are the BTB autograft and ST or ST/gracilis tendon autograft. Many clinical studies document good results with the use of BTB[3,11,14,25,41,44,45,54] and ST autografts.[1,11,41,44] Although several prospective, randomized, short-term and medium-term studies have been published,[2,6,13,16-18,30,44,48] to our knowledge, few reports on the long-term outcome of randomized controlled studies can be found in the literature.[29,42]

In the present study, good results were obtained in the majority of patients when using either type of graft in terms of the Lysholm score, Tegner activity level, and IKDC evaluation system, which is in line with the results of previous studies with shorter follow-up times.[13,14,16,34] We were unable to detect any significant differences between the BTB group and the ST group in terms of the functional outcome at 86 months. Several previous studies have reported anterior knee pain, disturbances in knee sensitivity, and kneeling discomfort using the BTB graft.[11,13,17,35,48] However, we were unable to demonstrate any basic difference in terms of donor-site morbidity at 7 years, which is contrary to the findings in several short-term studies. We speculate that, in the long term, anterior knee pain decreases for BTB patients and may also increase for ST patients. One of the reasons for this might be aging itself, together with the development of degenerative changes in the knee. Another reason might be that patients operated on using the BTB graft experience a reduction in their loss of sensitivity, or get used to it during such a long period of time. One must also consider that during harvest of the hamstring tendon, there is an increased risk of jeopardizing the saphenous nerve as well. In the long-term, this difference may also affect donor-site problems. The fact that there was no significant difference in donor-site morbidity in the long term is an important finding and might favor use of BTB grafts. Moreover, we were not able to demonstrate any significant differences between the groups in terms of loss of extension or flexion, which has been previously suggested as a frequent cause of anterior knee pain.[50]

Kartus et al.[33] reported that 65% of the patients were unable to perform or had difficulty performing the knee-walking test 2 years after patellar tendon harvest using a central vertical 7- to 8-cm incision. The corresponding value after using a 2-incision technique similar to the technique used in the present study was 47%, which is higher than in the present study.[33] This might indicate that kneeling and knee-walking problems decrease in the long term. Furthermore, in the present study, the clinical assessments did not reveal any differences in anterior knee laxity using the KT-1000 arthrometer or manual Lachman test, which is in line with some previous studies.[30,46] Contrary to these findings, other previous short- and medium-term studies have reported a better objective outcome in terms of laxity for the BTB graft compared with the ST graft.[2,6]

The decrease in laxity between the preoperative values and the values at follow-up as measured with the KT-1000 arthrometer did not reach statistical significance in the ST group. The reason for this might be too-small study groups or the distribution with a wide range in the measurements within the groups. One important feature of the present study is that the same fixation technique was used for both types of graft. However, concern has been raised about the use of interference screws for the fixation of soft tissue grafts on the tibial side.[40] Clinically, this does not appear to be a problem. The forces acting on the graft complex are estimated to be significantly less than 400 N when permitting full weightbearing and closed kinetic chain exercises without loaded terminal extension.[19,51] The forces are thereby under the pull-out limits for most fixation techniques. The ACL reconstruction does not result in normal knee function but gives the patient a chance to return to sporting activities, although usually at a lower level than before the injury. The preinjury Tegner activity level in this study was a median of 9 in both groups (eg, competitive sports such as soccer, gymnastics, ice hockey, or wrestling) and the median age was about 28 years, which is a typical patient population. At 7 years, the activity level was reduced by 3 to 4 levels compared with the preinjury level. Aging itself, as well as new interests in life, are certainly also responsible for this reduction.

The majority of the patients in the present study had meniscal injuries, which may influence the results. However, the number of meniscal injuries was evenly distributed, thereby not affecting the comparison between the 2 study groups.

CONCLUSION

On the basis of the present study, we conclude that the results were acceptable using both types of graft at 7 years after surgery. No clear advantage for either technique was demonstrated. Both techniques are reliable when it comes to improving patient performance, allowing a return to a higher level of activity than before surgery, and are therefore equally valid choices for ACL reconstructions even in the long term. Contrary to previous short-term studies, we found no significant differences in terms of donor-site morbidity and laxity between the study groups.

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